Barycentric Tether Calibration

HackerRank

You are the flight director on the Aegis deep space orbital platform. You have an array of N deployed satellites. Each satellite i (from 1 to N) has a structural mass of M[i] tons.

You need to connect pairs of satellites using carbon nanotube tethers to form communication relays. To prevent the tether from snapping under orbital tension, the pair must generate an Effective Anchor Mass of exactly S tons.

The Barycentric Rule: The tether system dynamically compensates for the mass difference between the two linked satellites. Because of centrifugal physics, the lighter satellite of the pair must spin its counter-thrusters twice as hard to maintain the center of gravity.

Mathematically, the Effective Anchor Mass of a tethered pair of satellites with masses A and B is evaluated as: 2 * min(A, B) + max(A, B)

Find the total number of unordered valid pairs of distinct satellites that yield an Effective Anchor Mass of exactly S.

Note: • If the two satellites have the exact same mass, either one can act as the lighter anchor.

Input Format

The first line contains a integer, N, denoting the number of deployed satellites.

The second line contains a integer, S, denoting the target effective anchor mass.

Each line i of the N subsequent lines (where 0 ≤ i < N) contains a integer, M[i].

Constraints

1 <= N <= 10^5

1 <= S <= 10^9

1 <= M[i] <= 10^9

Output Format

The output contains an integer value

Sample Input 0

```
5
7
1
2
3
4
5
```

Sample Output 0

```
2
```

Explanation 0

The pairs (1, 5) and (2, 3) both yield the target mass of 7 because 2*1 + 5 = 7 and 2*2 + 3 = 7. This results in exactly 2 valid pairs.

Sample Input 1

```
4
15
5
5
5
5
```

Sample Output 1

```
6
```

Explanation 1

Every pair of satellites with mass 5 satisfies the formula 2*5 + 5 = 15. Choosing any 2 of the 4 identical satellites gives 6 unique combinations.

Sample Input 2

```
3
10
1
2
3
```

Sample Output 2

```
0
```

Explanation 2

No pair of satellites from the masses 1, 2, and 3 can satisfy the formula to reach the target mass of 10. This results in 0 valid pairs.